

TOTAL QUALITY MANAGEMENT-23A03705

UNIT – I Introduction: Definition of Quality, Dimensions of Quality, Definition of Total quality management, Quality Planning, Quality costs – Analysis, Techniques for Quality costs, Basic concepts of Total Quality Management.

UNIT - II Historical Review: Quality council, Quality statements, Strategic Planning, Deming Philosophy, Barriers of TQM Implementation, Benefits of TQM, Characteristics of successful quality leader, Contributions of Gurus of TQM, Case studies.

UNIT – III TQM Principles: Customer Satisfaction – Customer Perception of Quality, Customer Complaints, Service Quality, Customer Retention, Employee Involvement – Motivation, Empowerment teams, Continuous Process Improvement – Juran Trilogy, PDCA Cycle, Kaizen, Supplier Partnership – Partnering, sourcing, Supplier Selection, Supplier Rating, Relationship Development, Performance Measures Basic Concepts, Strategy, Performance Measure Case studies.

UNIT - IV TQM Tools: Benchmarking – Reasons to Benchmark, Benchmarking Process, Quality Function Deployment (QFD) – House of Quality, QFD Process, Benefits, Taguchi Quality Loss Function, Total Productive Maintenance (TPM) – Concept, Improvement Needs, FMEA – Stages of FMEA, The seven tools of quality, Process capability, Concept of Six Sigma, New Seven management tools, Case studies.

UNIT – V Quality Systems: Need for ISO 9000 and Other Quality Systems, ISO 9000: 2000 Quality System – Elements, Implementation of Quality System, Documentation, Quality Auditing, QS 9000, ISO 14000 – Concept, Requirements and Benefits, Case Studies.

INTRODUCTION

Total Quality Management (TQM) is a management strategy aimed at embedding awareness of quality in all organizational processes. TQM has been widely used in manufacturing, education, government, and service industries, as well as space and science programs.

1 DEFINITION OF QUALITY:

- Predictable degree of uniformity and dependability at low cost and suited to the market
-**Deming**
- Fitness for use-**Juran**
- Conformance to requirements - **Crosby**
- Minimum loss imparted by a product to society from the time the product is shipped -
Taguchi
- A way of managing the organization -**Feigenbaum**
- Correcting and preventing loss, not living with loss - **Hosffin** .
- The totality of characteristics of an entity that bear on its ability to satisfy stated and implied needs – **ISO**

TQM is composed of three paradigms:

Total : Involving the entire organization, supply chain, and/or product life cycle.

Quality : Degree of Excellence of a product or service provide.

Management : The system of managing with steps like Plan, Organize, Control, Lead, Staff, provisioning.

QUANTIFICATION OF QUALITY

$$Q = P / E$$

P = Performance

E = Expectations

Q = Quality

2. DIMENSION OF QUALITY

Quality has 2 dimensions. These dimensions are product and service quality.

DIMENSION OF QUALITY	
Product quality	Service quality
Performance	Time
Features	Timeliness
Usability	Completeness
Conformance	Consistency
Reliability	Accessibility/Convenience
Durability	Accuracy
Maintainability/Serviceability	Responsiveness
Efficiency	Courtesy
Aesthetics	Competency/Expertise
Reputation	
Safety	

Dimensions of product quality:

1. **Performance:** primary product characteristics, e.g. picture brightness in TV.
2. **Features:** secondary characteristics, added features, e.g. remote control, picture-in-picture.
3. **Usability:** ease of use with minimum training.
4. **Conformance:** meeting specifications, industry standards,. (E.g. ISI specs., emission norms)
5. **Reliability:** consistency of performance over a specified time period under specified conditions.
6. **Durability:** extent of useful life, sturdiness.
7. **Maintainability/Serviceability:** ease of attending to maintenance, repairs.
8. **Efficiency:** ratio of output to input. E.g. mileage, braking distance, processing time.
9. **Aesthetics:** sensory characteristics, e.g. appearance, exterior finish, texture, color, shape, etc.
10. **Reputation:** subjective assessment based of past performance, brand image, industry ranking.
11. **Safety:** in items like pressure cookers, electrical items, toys, cranes, etc.

Dimensions of service quality:

1. **Time:** how much time a customer must wait / undergo service.
2. **Timeliness:** whether service will be performed when promised.
3. **Completeness:** whether all items in the order are included.
4. **Consistency:** consistent service every time, and for every customer.
5. **Accessibility/Convenience:** ease of obtaining the service.
6. **Accuracy:** absence of mistakes.
7. **Responsiveness:** quick response, resolution of unexpected problems.
8. **Courtesy:** cheerful, friendly service.
9. **Competency/Expertise:** In professions like doctors, lawyers, mechanics, etc.

3. DEFINITION OF TQM:

According to International Organization for Standardization (ISO):

“TQM is a management approach for an organization, centred on quality, based on the participation of all its members and aiming at long-term success through customer satisfaction, and benefits to all members of the organization and to society.”

TQM is an integrated organizational approach in delighting customers (both internal and external) by meeting their expectations on a continuous basis through everyone involved with the organization working on continuous improvement in all products, services, and processes along with proper problem solving methodology.

TQM is people - focused management system that aims at continual increase in customer satisfaction at continually lower cost. TQM is a total system approach (not a separate area of program), and an integral part of high level strategy. It works horizontally across functions and departments, involving all employees, top to bottom, and exceeds backwards and forward to include the supply chain and the customer chain.

Origin and Evolution of Total Quality Management

The concept of Quality Control emerged as a distinct discipline in the United States in 1920 when statistical theory was first applied to product quality control.

The philosophy of TQM was developed in Japan in 1960's by two academicians Dr. W. Edward Deming and Dr. J.M. Juran which involves total participation of all the employees at every level of the organization in quality management and not only the quality control staff.

In the 1980s to the 1990s, a new phase of quality control and management began. This became known as Total Quality Management (TQM). Having observed Japan's success of employing quality issues, western companies started to introduce their own quality initiatives. A typical definition of TQM includes phrases such as: customer focus, the involvement of all employees, continuous improvement and the integration of quality management into the total organization

Elements of TQM

- ✓ A sustained management commitment to quality.
- ✓ Total Focus on the Customer.
- ✓ Preventing rather than detecting Defects.
- ✓ Universal Quality Responsibility
- ✓ Quality Measurement
- ✓ Continuous Improvement
- ✓ Root Cause Corrective Actions
- ✓ Employees Involvement and Empowerment
- ✓ Focus upon Team Synergy
- ✓ Thinking Statistically
- ✓ Benchmarking
- ✓ Training
- ✓ Supplier Teaming

Objectives of TQM

- ✓ TQM emphasizes upon collective effort of all functional department and people for improvement in quality of goods and services in order to achieve higher customer satisfaction.
- ✓ The aim of TQM is to look for maximum satisfaction to the consumer by providing goods and services which are best in quality (i.e zero defects)
- ✓ TQM aims at educating and training the managers and employees since they are considered to be the integral part of the TQM process.

- ✓ TQM not only focus upon quality but also on productivity as it aims for Zero defect production which not only makes employees responsible for quality improvement but also leads to higher productivity.
- ✓ TQM aims at enhanced communication in the organization as every employee is encouraged to express their suggestion for quality improvement, cost reduction and elimination of wastage. It also calls for rewarding those who have active participation.

Advantage of TQM

- ✓ Enhanced Customer Satisfaction
- ✓ Total change in organization working culture
- ✓ Increased Productivity and efficiency
- ✓ Incorporation of advanced production techniques
- ✓ Development of new products and skills
- ✓ Enhanced teamwork
- ✓ Reduced rework
- ✓ Reduced Inventory
- ✓ Increased Profitability

Limitation of TQM

- ✓ TQM is a slow moving process. It requires total change in the outlook of management and employees. It's benefit will be therefore available after a longer period of time
- ✓ The success of TQM largely depends upon participative management. TQM needs employees who can take leads whereas trade unions are more interested in their own benefit rather than quality management
- ✓ TQM implementation is not an easy task particularly in a developing economy due to unfavorable approach of management and employees.

4. QUALITY PLANNING:

Before offering any product/service to the market, organizations need to plan upfront, about the level of quality, the features and the services they are going to offer in the market. They need to decide the customer segment they are going to serve with the products which they are making.

The steps for quality planning are as follows:

- **Identify who are the customers:** If a company is making a car, it has to decide which segment of the society it is going to serve. Appropriately requirements have to be met.
- **Determine the customer needs:** An in-depth survey and detailed analysis has to be done to know the requirements of the customer. Customers may ask for an automobile but their actual need to be collected to know whether they are giving value for aesthetics or reliability.
- **Translate the needs into language:** Once requirements are noted, then the data has to be analyzed to know the pattern or trend. Accordingly, the features and performance of the product have to be defined.
- **Develop a product to meet the needs:** Once requirements have been translated to needs and parameters, a prototype need to be produced.
- **Optimize the product so as to meet the company's as well as the customers' needs:** Matching the customers' requirement with the organizations capability to produce, it could be decided what features need to be added to the product and how the same could be manufactured. This is done in an organized process by making use of advanced tools and techniques, such as Quality Function Deployment and Failure Mode Effect Analysis.
- **Develop the Process to produce the product:** At this stage the knowledge and record in the earlier step is transferred to technical requirements and the machinery and processes need to produce are added to the organization.
- **Optimize the Process:** The process has to be fine tuned to optimize the working condition in order to minimize wastage and delay.
- **Prove the processes can a make the product under optimal conditions:** Produce products in the setup to ensure that the setting of the process results in cost effective products and improves efficiency of operation.

5. QUALITY COSTS

Definition:

Quality Cost is the sum of costs incurred to prevent defects, evaluate product quality, and correct defects before and after delivery to customers.

Components of Quality Cost:

Quality costs are classified into **four categories:**

1. Prevention Costs

These are costs incurred to **prevent defects** from occurring during production or service delivery.

Examples

- Quality planning
- Employee training
- Supplier evaluation
- Process improvement
- Preventive maintenance
- Product design reviews
- Calibration of equipment

Advantages

- Reduces defects
- Improves productivity
- Lowers overall quality cost

2. Appraisal Costs

These are costs incurred to **measure, inspect, and evaluate** products or services to ensure they meet quality standards.

Examples

- Inspection of raw materials
- Testing finished products
- Quality audits
- Laboratory testing
- Equipment calibration
- Process monitoring

Purpose

- Detect defects before products reach customers.

3. Internal Failure Costs

These costs occur when defects are found **before the product reaches the customer**.

Examples

- Scrap
- Rework
- Re-inspection
- Machine downtime due to defects
- Process failures

- Material wastage

Effects

- Increased production cost
- Reduced efficiency
- Delay in delivery

4. External Failure Costs

These costs occur when defective products reach the **customer**.

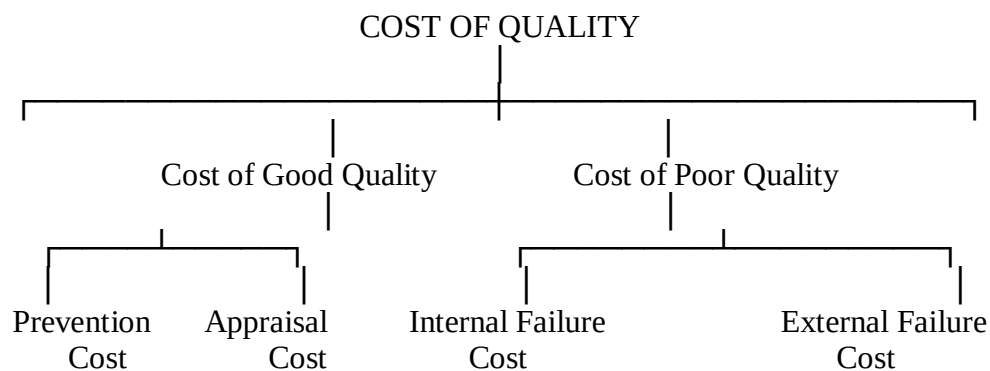
Examples

- Warranty claims
- Product replacement
- Product recalls
- Complaint handling
- Legal liabilities
- Loss of customer goodwill
- Loss of future sales

Effects

- Customer dissatisfaction
- Damage to company reputation
- Financial loss

Classification Diagram



Formula

Cost of Quality=Prevention Cost+Appraisal Cost+Internal Failure Cost+External Failure Cost

Example

A company spends:

- Prevention Cost = ₹50,000
- Appraisal Cost = ₹30,000
- Internal Failure Cost = ₹20,000
- External Failure Cost = ₹10,000

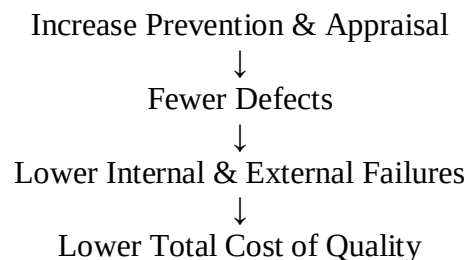
Total Cost of Quality = ₹1,10,000

Importance of Quality Cost in TQM

- Helps identify quality-related expenses.
- Reduces defects and waste.
- Improves customer satisfaction.
- Increases profitability.
- Supports continuous improvement.
- Enhances product reliability.
- Improves organizational efficiency.
- Helps management make informed decisions.

Relationship Between Quality Costs

- Increasing **prevention** and **appraisal** costs generally **reduces internal** and **external failure** costs.
- Investing in prevention is usually more economical than paying for failures after defects occur.



Advantages of Managing Quality Costs

- Better product quality
- Reduced manufacturing costs
- Lower warranty expenses
- Improved customer loyalty
- Increased productivity

- Better use of resources
- Competitive advantage
- Higher profits

Disadvantages

- Requires investment in training and quality systems.
- Initial implementation costs may be high.
- Continuous monitoring and documentation are needed.
- Benefits may take time to become visible.

ANALYSIS OF QUALITY COSTS:

1. Trend analysis
2. Pareto analysis

ANALYSIS TECHNIQUES OF QUALITY COST

The purpose of quality cost analysis is to determine the cost of maintaining a certain level of quality. Such activity is necessary to provide feedback to management on the performance of quality assurance and to assist management in identifying opportunities.

INDEX NUMBERS:

Index Numbers are often used in a variety of applications to measure prices, costs (or) other numerical quantities and to aid managers in understanding how conditions in one period compare with those in other periods.

A simple type of index is called a relative index

QUARTER	COST IN RS.	QUARTER	COST RELATIVE INDEX
1	2000	1	$(2000/2000) \times 100 = 100$
2	2200	2	$(2200/2000) \times 100 = 110$
3	2100	3	$(2100/2000) \times 100 = 105$
4	1900	4	$(1900/2000) \times 100 = 95$

Table No.1 Simple Index

Table No.2 Cost Relative Index

Cost Index in quarter t = (Cost in quarter t / Base period cost) x 100

TREND ANALYSIS:

- Good visual aids are important communication tools.

- Graphs are particularly useful in presenting comparative results to management Trend Analysis is one where Time-to-Time comparisons can be made which illustrates

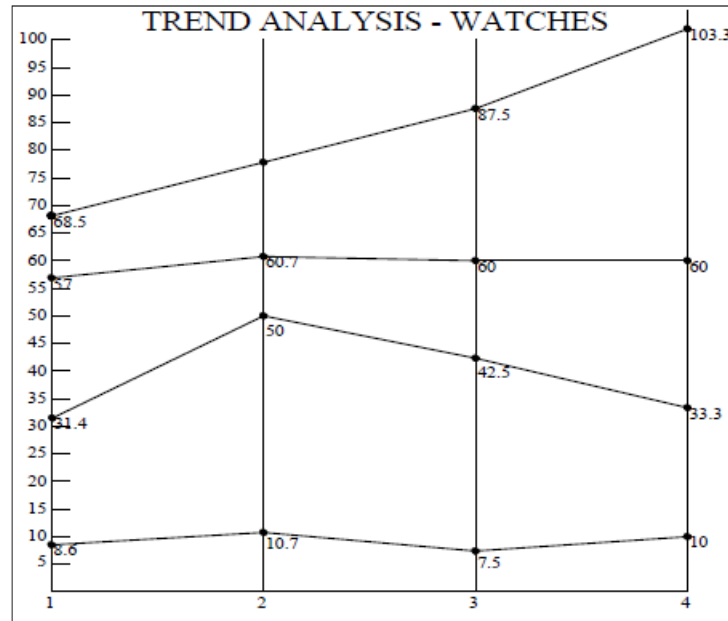


Figure: Trend Analysis –Watches

PARETO ANALYSIS:

Joseph Juran observed that most of the quality problems are generally created by only a few causes. For example, 80% of all internal failures are due to one (or) two manufacturing problems. Identifying these “vital few” and ignoring the “trivial many” will make the corrective action give a high return for a low money input.

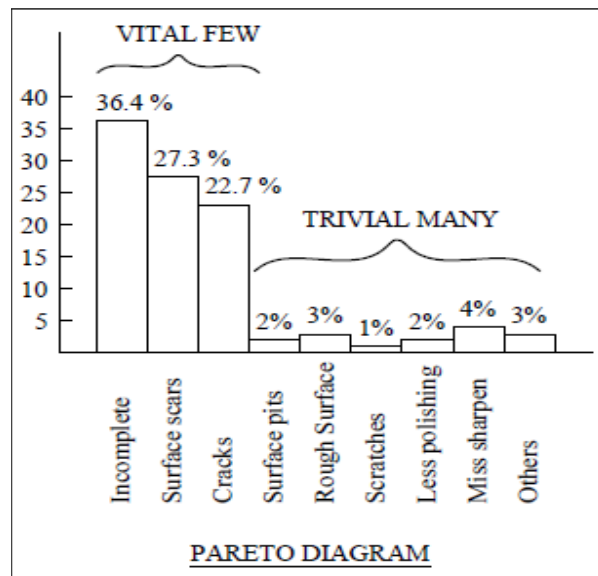


Figure: Pareto Diagram

Basic Concepts (Principles) of Total Quality Management (TQM)

Definition

Total Quality Management (TQM) is a management philosophy that focuses on continuously improving the quality of products, services, processes, and organizational performance by involving every employee and satisfying customer needs.

1. Customer Focus

Definition

Customer focus means understanding the needs and expectations of customers and providing products or services that satisfy or exceed those expectations. In TQM, the customer is considered the most important person because the success of an organization depends on customer satisfaction.

Key Features

- Understand customer needs.
- Deliver quality products and services.
- Collect customer feedback.
- Resolve customer complaints promptly.
- Build long-term customer relationships.

Importance

- Improves customer loyalty.
- Increases sales and profits.
- Enhances company reputation.
- Helps retain existing customers.

Example

A mobile phone company regularly updates its software based on customer feedback to improve user experience.

2. Leadership

Definition

Leadership means that top management provides a clear vision, direction, and commitment toward achieving quality objectives. Leaders create an environment where employees work together to improve quality continuously.

Responsibilities of Leaders

- Develop quality policies.
- Set quality objectives.
- Allocate resources.
- Encourage teamwork.
- Motivate employees.
- Promote ethical practices.

Importance

- Creates a quality culture.
- Motivates employees.
- Ensures successful implementation of TQM.
- Improves organizational performance.

Example

A factory manager conducts weekly quality review meetings and encourages employees to suggest improvements.

3. Employee Involvement

Definition

Employee involvement means every employee participates in quality improvement activities. Quality is everyone's responsibility, not only the quality department's.

Methods of Employee Involvement

- Teamwork
- Quality Circles
- Suggestion schemes
- Training programs
- Cross-functional teams

Benefits

- Higher employee motivation.
- Better problem-solving.
- Increased productivity.
- Improved communication.
- Greater innovation.

Example

Employees suggest ways to reduce material waste, leading to lower production costs.

4. Process Approach

Definition

A process is a set of related activities that transforms inputs into outputs. The process approach means managing activities as interconnected processes rather than individual tasks.

Process Cycle

Input → Process → Output → Customer Feedback → Improvement

Advantages

- Reduces errors.
- Improves efficiency.
- Standardizes work.
- Better quality control.
- Easier monitoring.

Example

In a bakery:

- Flour, sugar, butter → Mixing → Baking → Packaging → Customer

Each step is monitored to ensure consistent quality.

5. Continuous Improvement (Kaizen)

Definition

Continuous improvement means constantly making small improvements in products, services, and processes instead of waiting for major changes.

The Japanese word Kaizen means "Change for Better."

PDCA Cycle (Deming Cycle)



Benefits

- Reduces defects.
- Lowers production costs.
- Improves efficiency.
- Increases customer satisfaction.
- Encourages innovation.

Example

Reducing machine setup time from 30 minutes to 20 minutes through regular improvements.

6. Fact-Based Decision Making

Definition

Decisions should be based on facts, measurements, and data rather than assumptions or personal opinions.

Tools Used

- Pareto Chart
- Histogram
- Control Chart
- Scatter Diagram
- Cause-and-Effect Diagram
- Statistical Process Control (SPC)

Importance

- Accurate decisions.
- Reduces risks.
- Improves process control.
- Helps identify root causes.

Example

A company analyzes defect data and finds that 80% of defects come from one machine. Repairing that machine significantly improves product quality.

7. Integrated System

Definition

An integrated system means all departments in an organization work together toward common quality objectives. Every department contributes to customer satisfaction.

Departments Involved

- Production
- Purchasing
- Marketing
- Finance
- Human Resources
- Quality Control

Importance

- Better coordination.
- Faster communication.
- Improved efficiency.
- Consistent quality.

Example



Every department works together to achieve the same goal.

8. Strategic and Systematic Approach

Definition

Quality should be part of the organization's long-term strategic planning. Improvement activities should follow a planned and systematic approach.

Activities

- Set quality goals.
- Develop quality policies.
- Allocate resources.
- Monitor performance.
- Review results regularly.

Importance

- Achieves long-term success.

- Supports continuous improvement.
- Aligns quality with business objectives.
- Reduces operational risks.

Example

A company sets a five-year goal to reduce product defects by 50% through employee training and automation.

9. Effective Communication

Definition

Communication is the exchange of information between management, employees, suppliers, and customers. Good communication ensures everyone understands quality goals and responsibilities.

Types of Communication

- Upward communication
- Downward communication
- Horizontal communication

Importance

- Improves teamwork.
- Prevents misunderstandings.
- Speeds up problem-solving.
- Encourages employee participation.

Example

Daily production meetings help workers identify and resolve quality issues before they affect production.

10. Supplier Relationship Management

Definition

Organizations should build strong relationships with suppliers to ensure a continuous supply of high-quality raw materials and services. Good supplier relationships improve product quality and reduce production problems.

Activities

- Supplier selection
- Supplier evaluation

- Long-term partnerships
- Quality audits
- Performance monitoring

Importance

- Better raw material quality.
- Reduced production delays.
- Lower costs.
- Improved product reliability.
- Strong business relationships.

Example

An automobile manufacturer works closely with its tyre supplier to ensure consistent quality, reducing defects and customer complaints.

Diagram of TQM Principles



Advantages of TQM Principles

- Improves product and service quality.
- Increases customer satisfaction.
- Reduces defects and waste.
- Lowers production costs.

- Improves teamwork.
- Enhances employee morale.
- Supports continuous improvement.
- Strengthens organizational competitiveness.
- Increases profitability.
- Builds long-term business success.